CRITICAL CARE NURSING · LECTURE 2 · QUESTION BANK

Acid-Base Balance & its Disorders

توازن الحمض والقاعدة واضطراباته — بنك الأسئلة

98	6	12
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

ميدترم كريتكال 2022 · فاينال 2021/2022 ترم أول (صور الجروب) · فاينال 2021/2022 ترم تاني (صور الجروب) · أسئلة سابقة · كويز 1 كريتكال – ترم أول 2024 · كويز 1 كريتكال – ترم أول 2023 · ميدترم كريتكال 2021

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 فاينال 2021/2022 ترم تاني (صور الجروب)

You would interpret this ARDS patient's ABG (pH 7.31, PaCO2 68, HCO3 26) as:

a. Uncompensated metabolic alkalosis
b. Uncompensated respiratory acidosis
c. Partially compensated metabolic acidosis
d. Partially compensated respiratory acidosis

2 فاينال 2021/2022 ترم تاني (صور الجروب)

To ensure safety of a patient with severe metabolic alkalosis, the nurse should anticipate and monitor for: (1) GIT disturbances (2) shallow breathing (3) flaccid hypotonia (4) tetany & dysrhythmias

a. 1,2,3
b. 1,2,4
c. 1,3,4
d. 2,3,4

3 فاينال 2021/2022 ترم أول (صور الجروب)

An elderly patient whose ABG: pH 7.1, PaCO2 (value missing) mmHg, HCO3 10 mmol/L is expected to have partially compensated:

a. Respiratory acidosis
b. Respiratory alkalosis
c. Metabolic alkalosis
d. Metabolic acidosis

4 فاينال 2021/2022 ترم أول (صور الجروب)

Which ABG result is expected in a patient admitted to ICU with atelectasis?

a. pH 7.25, PaCO2 50 mmHg
b. pH 7.35, PaCO2 40 mmHg
c. pH 7.50, PaCO2 52 mmHg
d. pH 7.47, PaCO2 28 mmHg

5 When ABG values are pH 7.48, PaCO2 30 mmHg, HCO3 15 mEq/L the patient will have: فاينال 2021/2022 ترم أول (صور الجروب)

a. Partially compensated metabolic acidosis
b. Partially compensated metabolic alkalosis
c. Partially compensated respiratory alkalosis
d. Partially compensated respiratory acidosis

6 A patient with high anion gap metabolic acidosis due to DKA is expected to have which sign? ميدترم كريتكال 2022

a. Hypoventilation
b. Neuro-excitability
c. Kussmaul's respiration
d. Blood glucose of 110 mg/dL

7 A patient whose pH 7.23, PaCO2 33 mmHg, HCO3 15 mEq/L is expected to have: ميدترم كريتكال 2022

a. Uncompensated respiratory alkalosis
b. Partially compensated metabolic acidosis
c. Partially compensated metabolic alkalosis
d. Partially compensated respiratory acidosis

8 A patient with atelectasis is expected to have which ABG findings? ميدترم كريتكال 2022

a. pH 7.25, PaCO2 50 mmHg
b. pH 7.35, PaCO2 40 mmHg
c. pH 7.50, PaCO2 52 mmHg
d. pH 7.47, PaCO2 28 mmHg

9 A patient's ABG showed pH 6.90, HCO3 14 mmol/L, PaCO2 68 mmHg. The patient is expected to have: ميدترم كريتكال 2021

a. Partially compensated metabolic acidosis
b. Partially compensated respiratory acidosis
c. Simultaneous respiratory and metabolic acidosis
d. Respiratory acidosis with complete renal compensation

10 When pH is 7.48, PaCO2 30 mmHg and HCO3 15 mEq/L, the patient is expected to have: ميدترم كريتكال 2021

a. Partially compensated metabolic alkalosis
b. Partially compensated respiratory alkalosis
c. Partially compensated metabolic acidosis
d. Partially compensated respiratory acidosis

11 When ABG values are pH 7.1, PaCO2 16 mmHg and HCO3 10 mmol/L, the patient is expected to have: ميدترم كريتكال 2021

a. Partially compensated respiratory acidosis
b. Partially compensated respiratory alkalosis
c. Partially compensated metabolic alkalosis
d. Partially compensated metabolic acidosis

12 Which of the following is an expected ABG finding in a patient with atelectasis? ميدترم كريتكال 2021

تكرر في 2 مصادر

a. pH 7.25, PaCO2 50 mmHg
b. pH 7.35, PaCO2 40 mmHg
c. pH 7.50, PaCO2 52 mmHg
d. pH 7.47, PaCO2 28 mmHg

13 A patient with ABG values of pH 7.39, PaCO2 27 mmHg and HCO3 19 mmol/L is expected to have: كويز 1 كريتكال – ترم أول 2023

a. Uncompensated metabolic acidosis
b. Uncompensated respiratory acidosis
c. Fully compensated metabolic acidosis
d. Fully compensated respiratory alkalosis

14 The expected finding for a young woman found comatose with ABG pH 6.90, HCO3 13 mmol/L and PaCO2 68 mmHg is: كويز 1 كريتكال – ترم أول 2023

a. Uncompensated respiratory acidosis
b. Fully compensated respiratory acidosis
c. Partially compensated metabolic acidosis
d. Mixed respiratory and metabolic acidosis

15 ABG for a 57-year-old patient: pH 7.21, Na 145, K 4.5, Cl 105, HCO3 25 mEq/L. The calculated anion gap is: كويز 1 كريتكال – ترم أول 2023

a. 12
b. 8
c. 10
d. 15

16 Which nursing measure is directed for management of a post car accident patient who has respiratory alkalosis? كويز 1 كريتكال – ترم أول 2023

a. Ask if he is alright
b. Administer O2 therapy
c. Check body temperature
d. Ask him to breathe into a bag

17 ABG for a postoperative patient who received IV morphine: pH 7.10, PaCO2 70 mmHg, HCO3 24 mEq/L indicates: كويز 1 كريتكال – ترم أول 2023

a. Uncompensated metabolic acidosis
b. Uncompensated respiratory acidosis
c. Partially compensated metabolic alkalosis
d. Partially compensated respiratory alkalosis

18 Mr. A, 54 years, admitted with SOB, pyrexia and productive cough. ABG: pH 7.3, PaCO2 68, HCO3 28, PaO2 60. How would you interpret this? كويز 1 كريتكال – ترم أول 2024

a. Respiratory acidosis, uncompensated
b. Respiratory acidosis, partially compensated
c. Metabolic alkalosis, uncompensated
d. Metabolic acidosis, partially compensated

19 **You would interpret ABG values of pH 7.20, PaO2 90, PaCO2 23 and HCO3 12 mmol/L as:** كويز 1 كريتكال – ترم أول 2024

a. Uncompensated metabolic alkalosis
b. Uncompensated respiratory acidosis
c. Partially compensated metabolic acidosis
d. Partially compensated respiratory acidosis

20 **When ABG results indicate pH 7.55, HCO3 22 mEq/L and PaCO2 30 mmHg, the patient is expected to have:** كويز 1 كريتكال – ترم أول 2024

a. Metabolic acidosis
b. Metabolic alkalosis
c. Respiratory acidosis
d. Respiratory alkalosis

21 **A 63-year-old drowsy patient on 10 L O2: pH 7.29, PaCO2 90, HCO3 26. Interpretation:** أسئلة سابقة

a. Uncompensated respiratory acidosis
b. Uncompensated metabolic acidosis
c. Partially compensated respiratory acidosis
d. Fully compensated respiratory acidosis

22 **A 17-year-old with chest tightness, SOB and tingling fingers: pH 7.49, PaCO2 32, HCO3 22. Interpretation:** أسئلة سابقة

a. Uncompensated respiratory alkalosis
b. Uncompensated metabolic alkalosis
c. Partially compensated respiratory alkalosis
d. Mixed alkalosis

23 **A 48-year-old with bowel obstruction and profuse vomiting: pH 7.50, PaCO2 55, HCO3 29. Interpretation:** أسئلة سابقة

a. Partially compensated metabolic alkalosis
b. Uncompensated respiratory acidosis
c. Partially compensated respiratory acidosis
d. Mixed alkalosis

24 **A 59-year-old short of breath on 3 L O2: pH 7.35, PaCO2 84, HCO3 29. Interpretation:** أسئلة سابقة

a. Fully compensated respiratory acidosis
b. Uncompensated respiratory acidosis
c. Fully compensated metabolic alkalosis
d. Partially compensated metabolic alkalosis

25 **A man found unconscious, RR 6, pinprick pupils (given naloxone): PaCO2 71, HCO3 22, pH 7.31. Interpretation:** أسئلة سابقة

a. Uncompensated respiratory acidosis
b. Partially compensated respiratory acidosis
c. Uncompensated metabolic acidosis
d. Mixed acidosis

26 After cardiac arrest and 6 minutes of CPR: HCO3 18, pH 7.14, PaCO2 55. Interpretation: أسئلة سابقة

a. Mixed respiratory and metabolic acidosis
b. Uncompensated metabolic acidosis
c. Partially compensated respiratory acidosis
d. Uncompensated respiratory acidosis

27 Diabetic ketoacidosis, aspirin toxicity, and renal failure are causes of : أسئلة سابقة

a. Normal anion gap metabolic acidosis
b. High anion gap metabolic acidosis
c. Low anion gap metabolic acidosis
d. Normal anion gap respiratory acidosis

28 The nurse interprets the ABG values of pH: 7.45, PaCO2; 30 mm Hg, and HCO3; 20 mmol/L as: أسئلة سابقة

a. Compensated Metabolic acidosis
b. Uncompensated Respiratory acidosis,
c. Uncompensated Metabolic alkalosis
d. Compensated Respiratory alkalosis

29 All are true for renal handling of acids in metabolic acidosis except: أسئلة سابقة

a. Hydrogen ion secretion is increased
b. Bicarbonate reabsorption is decreased
c. Urinary acidity is increased
d. Urinary ammonia is increased

30 A patient's ABG values revealed that: pH: 7.52, PaCO2: 48 mmHg and HCO3: 36 mmol/L, is expected to have ..... أسئلة سابقة

a. Uncompensated Metabolic acidosis
b. Fully Compensated Metabolic alkalosis
c. Partially Compensated Metabolic acidosis
d. Partially Compensated Metabolic alkalosis Abed Critical Theory

31 A patient was found unresponsive, his ABG indicated that: pH is 7.12, PaCO2 is 90 mmHg, and HCO3 is 22 mmol/L. The nurse interprets the results as indicating which condition ? أسئلة سابقة

a. Metabolic acidosis with compensation
b. Respiratory acidosis with compensation
c. Metabolic acidosis without compensation Respiratory acidosis without compensation
d.

32 The ABG analysis for a postoperative patient who received I.V. Morphine Sulfate showed that: PH; 7.10, PaCO2 70 mmHg, and :HCO3: 24 mEq/L indicate أسئلة سابقة

a. Uncompensated Metabolic Acidosis,
b. Uncompensated Respiratory Acidosis
c. Partially Compensated Metabolic Alkalosis
d. Partially Compensated Respiratory Alkalosis

33 The expected finding for a young woman who was found comatosed and her ABG values are: pH: 6.90, HCO3: 13 mmol/L, and PaCO2: 68 mmHg is .....

أسئلة سابقة

a. Uncompensated Respiratory Acidosis
b. Fully Compensated Respiratory Acidosis
c. Partially Compensated Metabolic Acidosis
d. Mixed Respiratory & Metabolic Acidosis

34 patient with ABG values of pH:7.39, PaCO2:27 mmHg and HCO3: 19 mmol/L, is expected to have

أسئلة سابقة

a. Uncompensated Metabolic Acidosis
b. Uncompensated Respiratory Acidosis
c. Fully Compensated Metabolic Acidosis Fully Compensated Respiratory Alkalosis
d. Abed Critical Theory

35 A client with a history of anxiety experiences respiratory alkalosis. The nurse measures a respiratory rate of 46 breaths/minute, a heart rate of 110 beats/minute, a blood pressure of 162/90 mm Hg, and a temperature of 98.6° F (37° C). To help correct respiratory alkalosis, the nurse should:

أسئلة سابقة

a. Administer antibiotics as ordered
b. Insert an NG tube as ordered
c. Administer acetaminophen as ordered
d. Instruct the client to breathe into a paper bag

36 A nurse is assigned patients with acid-base irregularities. While caring for a patient with gastric lavage, the nurse should assess for

أسئلة سابقة

a. Respiratory acidosis
b. Metabolic acidosis
c. Metabolic alkalosis
d. Respiratory alkalosis

37 A nurse is caring for a client who is on a mechanical ventilator. Blood gas results indicated a pH of 7.50 and a Pco2 of 30. The nurse has determines that the client is experience respiratory alkalosis. Which laboratory value would most likely be noted in this condition ?

أسئلة سابقة

a. Sodium level of 145 mEq/L
b. Potassium level of 3.0 mEq/L
c. Magnesium level of 2.0 mg/dL
d. Phosphorus level of 4.0 mg/dL Abed Critical Theory

38 A nurse caring for a client with an ileostomy understands that the client is most a risk for developing which acid-base disorder ?

أسئلة سابقة

a. Metabolic Acidosis Metabolic Alkalosis
b.
c. Respiratory Acidosis
d. Respiratory Alkalosis

39 Choose the incorrect statement about anion gap out of the followings: أسئلة سابقة

a. In lactic acidosis anion gap is increased
b. Anion gap is decreased in Hypercalcemia
c. Anion gap is decreased in Lithium toxicity
d. Anion gap is decreased in ketoacidosis

40 A patient complain of atelectasis, which ABG results are associated with the patient condition أسئلة سابقة

a. PH: 7.35 Paco2 40 mmhg
b. PH: 7.47 Paco2 28 mmhg
c. PH: 7.25 - Paco2 50 mmhg
d. PH: 7.50 - Paco2 52 mmhg

41 When the ABG results show that pH is 7.39, PaCO2 is 27 mmHg and HCO3 is 19 mEq/L. The patient is expected to have: أسئلة سابقة

a. Metabolic Acidosis, Uncompensated
b. Respiratory Acidosis, Uncompensated
c. Metabolic Acidosis, Fully Compensated
d. Respiratory Alkalosis, Fully Compensated Abed Critical Theory

42 An 80-year-old man had a bad cold. After two weeks he said, "It went in to my chest, I am feeling tightness in my chest, I am coughing, suffocated and unable to breathe!" What could be the possible reason ? أسئلة سابقة

a. Metabolic Acidosis
b. Metabolic Alkalosis
c. Respiratory Acidosis Respiratory Alkalosis
d.

43 Which out of the following conditions will not cause respiratory alkalosis ? أسئلة سابقة

a. Fever
b. Anxiety
c. Laryngeal obstruction
d. Salicylate toxicity

44 A person was admitted to hospital in a coma Analysis of the arterial blood gas the following values: PCO2 16 mmHg, HCO3-5 mmol and pH 7.1. What is the underlying acid-base disorder ? أسئلة سابقة

a. Metabolic acidosis
b. Respiratory alkalosis
c. Metabolic alkalosis
d. Respiratory acidosis Abed Critical Theory

45 90, HCO3-13 meq/liter, PaCO2 68 mmHg. This patient's acid-bane status is most accurately described as أسئلة سابقة

a. Uncompensated metabolic acidosis
b. Uncompensated respiratory acidosis Simultaneous respiratory and metabolic acidosis
c.
d. Respiratory acidosis with partial renal compensation

46 What are the Acceptable Sites for Arterial Puncture? أسئلة سابقة

a. Dorsalis Pedis
b. Femoral Artery
c. Brachial Artery
d. Radial Artery

47 Which of the following laboratory results below indicates partially compensated metabolic Alkalosis ? أسئلة سابقة

a. Low p CO2, normal bicarbonate and, high pH
b. Low p CO2, low bicarbonate, low pH
c. High p CO2, normal bicarbonate and, low pH
d. High pCO2, high bicarbonate and High pH Abed Critical Theory

48 Baby LR was rushed to the Emergency Room following her mother's complaint that the infant has been irritable, difficult to breastfeed and has had diarrhea for the past 3 days. The infant's respiratory rate is elevated and the fontanels are sunken. The Emergency Room physician orders ABGS after assessing the ABCs. The results from the ABG results show pH 7.39, PaCO2 27 mmHg and HCO3 19 mEq/L. What does this mean? أسئلة سابقة

a. Respiratory Alkalosis, Fully Compensated
b. Metabolic Acidosis, Uncompensated
c. Metabolic Acidosis, Fully Compensated
d. Respiratory Acidosis, Uncompensated

49 The pH of the body fluids is stabilized by buffer systems. Which of the following compounds is the most effective buffer system at physiological pH? أسئلة سابقة

a. Bicarbonate buffer
b. Phosphate buffer
c. Protein buffer
d. All of the above

50 Which of the following signs may be exhibited as a compensatory mechanism in a patient who has ABG values of: HCO3: 38, pH: 7.50,.and PaCO2: 50 ? أسئلة سابقة

a. Increased potassium level(hyperkalemia)
b. Decreased potassium level (hyporkalemia)
c. Hyperventilation (tachypnea)
d. Hypoventilation (bradypnea) Abed Critical Theory

51 Mr. A is suffering from persistent vomiting for two days now He appeared to be lethargic and weak and has myalgia. He is diagnosed as having gastroenteritis and dehydration. Measurement of ABG showed that pH 7.5, Pa02 85 mm Hg, PaCO2 40 mm Hg, and HCO3 34 mEq/L. You would expect Mr. A to have which of the following acid-base disorder ? أسئلة سابقة

a. Metabolic Alkalosis, Uncompensated
b. Respiratory Alkalosis, Uncompensated
c. Metabolic Alkalosis, Partially Compensated
d. Respiratory Acidosis, Partially Compensated

52 ..... is NOT a cause of metabolic acidosis أسئلة سابقة

a. Aspirin toxicity
b. Carbonic anhydrase inhibitors
c. ileostomy
d. Hyperaldosteronism

53 Which of the following is NOT a cause of metabolic alkalosis ? أسئلة سابقة

a. Usage of Diamox
b. Hyperaldosteronism
c. Nasogastric feeding
d. Diuretic therapy

54 A patient who's ABG analysis showed that: pH: 6.90, HCO3: 14 mmol/L, PaCO2: 68 mmHg, is expected to have: أسئلة سابقة

a. Partially compensated metabolic acidosis
b. Partially compensated respiratory acidosis
c. Simultaneous respiratory and metabolic acidosis. Respiratory acidosis with complete renal compensation
d. Abed Critical Theory

55 When the PH is 7.48, PaCo2: 30 mmHg, and HCO3; 15. mEq/l, the patient is expected to have which of the following acid base alterations ? أسئلة سابقة

a. Partially compensated metabolic alkalosis
b. Partially compensated respiratory alkalosis
c. Partially compensated metabolic acidosis
d. Partially compensated respiratory acidosis

56 When the ABG analysis values are: PH, 7.1, PaCo2, 16 mmHg, and HCO3, 10 mmol/l, the patient is expected to have أسئلة سابقة

a. Partially compensated respiratory acidosis
b. Partially compensated respiratory alkalosis
c. Partially compensated metabolic alkalosis
d. Partially compensated metabolic acidosis

57 Which of the following is an expected ABG finding in a patient with atelectasis ? أسئلة سابقة

تكرر في 2 مصادر

a. PH 7.25-PaCo2, 50 mmHg
b. PH 7.35-PaCo2, 40 mmHg
c. PH 7.50- PaCo2, 52 mmHg
d. PH: 7.47-PaCo2, 28 mmHg

58 Which of the following is the expected acid-base imbalance for a patient who was on gastric suctioning for 4 days, and blood gasses values of pH; 7.50, PaCO2; 45 mm Hg, HCO3: 32 mmol/l أسئلة سابقة

a. Compensated Metabolic Alkalosis
b. Uncompensated Metabolic Alkalosis
c. Compensated Respiratory Alkalosis
d. Uncompensated respiratory Alkalosis. Abed Critical Theory

59 A mountaineer attempts an assault on a high mountain in the Andes and reaches an altitude of 5000 meters (16,400 ft) above sea level. What will happen to his arterial PCO2 and pH? أسئلة سابقة

a. Both will be lower than normal
b. Both will be higher than normal
c. The pH will rise and PCO2 will fall. The pH will fall and PCO2 will rise
d.

60 55, PCO2: 45 mmHg, and HCO3: 40 mmol/l ? أسئلة سابقة

a. Metabolic Acidosis
b. Metabolic Alkalosis
c. Respiratory Acidosis
d. Respiratory Alkalosis

61 The nurse is caring for a client having respiratory distress related to an anxiety attack. Recent arterial blood gas values are pH = 7.53, Pao2 = 72 mm Hg (72 mm Hg), Paco2 = 32 mm Hg (32 mm Hg), and HCO3- = 28 mEq/L (28 mmol/L). Which conclusion about the client should the nurse make? أسئلة سابقة

a. The client has acidotic blood
b. The client is probably overreacting
c. The client is fluid volume overloaded
d. The client is probably hyperventilating. Abed Critical Theory

62 A patient who's ABG analysis showed that: pH: 7.48. PaO2: 83 mm Hg, PaCO2: 38 mmHg, and HCO3: 34 mmol/L is expected to have: أسئلة سابقة

a. Uncompensated Metabolic Alkalosis,
b. Uncompensated Respiratory Alkalosis, Compensated
c. Partially Metabolic Alkalosis,
d. Partially Compensated Respiratory Acidosis

63 When the serum Na: 145 mEq/L, CL: 105 mEq/L, HCO3: 21 mEq/L, the patient's calculated anion gap will be: أسئلة سابقة

a.
b. 8
c. 11
d.

64 A patient who's ABG analysis showed that: pH: 7.47, PaO2: 85 mm Hg, PaCO2: 40 mmHg, and HCO3: 34 mmol/L is expected to have: أسئلة سابقة

a. Uncompensated Metabolic Alkalosis ,
b. Uncompensated Respiratory Alkalosis,
c. Partially Compensated Metabolic Alkalosis,
d. Partially Compensated Respiratory Acidosis

65 When the serum Na: 142 mEq/L, CL: 107 mEq/L, HCO3: 23 mEq/L, the patient's calculated anion gap will be: أسئلة سابقة

a. 13
b.
c. 12
d. 18 Abed Critical Theory

66 Which of the following is the expected acid-base imbalance for a patient who was on gastric suctioning for 3 days, and blood gasses values of: pH; 7.48, PaCO2; 45 mm Hg, HCO3: 29 mmol/l أسئلة سابقة

a. Compensated Metabolic Alkalosis
b. Uncompensated Metabolic Alkalosis
c. Compensated Respiratory Alkalosis. Uncompensated respiratory Alkalosis
d.

67 أسئلة سابقة

A patient with ABG values of pH: 7.38, PaCO2: 47 mmHg and HCO3: 30 mmol/L, is expected to have:

a. Uncompensated Metabolic Acidosis
b. Uncompensated Respiratory Acidosis
c. Fully Compensated Metabolic Alkalosis
d. Fully Compensated Respiratory Acidosis

68 أسئلة سابقة

50 and a Paco2 of 30 mm Hg (30 mm Hg). The nurse monitors the client for which clinical manifestations associated with these ABG results? Select all that apply

a. Nausea
b. Confusion
c. Papilledema
d. Lightheadedness
e. 1,2,4
f. 1.2.3
g. 1,3,4
h. 2,3,4 Abed Critical Theory

69 أسئلة سابقة

The ABG analysis of a patient with acute respiratory distress syndrome (ARDS) showed that pH: 7.31, PaCO2 68 mmHg, HCO3: 26 mmol/L, and PaO2 60 mmHg, 02 saturation: 80%. The patient was intubated, mechanically ventilated, and the physician recommended keeping him in prone position. You would interpret the above patient's ABG result as:

a. Uncompensated Metabolic Alkalosis
b. Uncompensated Respiratory Acidosis
c. Partially Compensated Metabolic Acidosis
d. Partially Compensated Respiratory Acidosis

70 أسئلة سابقة

Mr. A is 54-year-old admitted to the ICU with increasing shortness of breath, pyrexia, and a productive cough with yellow-green sputum. arterial blood gas shows pH 7.3, PaCO2 68 mm Hg, HCO3 28 mmol/L, and PaO2 60 mm Hg. How would you interpret this ?

a. Respiratory Acidosis, Uncompensated
b. Respiratory Acidosis, Partially Compensated
c. Metabolic Alkalosis, Uncompensated
d. Metabolic Acidosis, Partially, Compensated

71 أسئلة سابقة

Which of the following acid-base alterations may be expected in a patient with septic shock ?

a. Metabolic acidosis
b. Metabolic alkalosis
c. Respiratory acidosis
d. Respiratory alkalosis. Abed Critical Theory

72 أسئلة سابقة

You would interpret ABG values of PH: 7.20, PaO2: 90 mmHg, PaCO2: 23 mmHg, and HCO3: 12 mmol/L ?

a. Uncompensated Metabolic Alkalosis
b. Uncompensated Respiratory Acidosis
c. Partially Compensated Metabolic Acidosis
d. Partially Compensated Respiratory Acidosis

73 When arterial blood gas (ABG) results indicate a pH of 7.55, HCO3 of 22mEq/l and a PaCO2 mm of 30 mmHg, the patient is expected to: أسئلة سابقة

a. Metabolic acidosis
b. Metabolic alkalosis
c. Respiratory acidosis
d. Respiratory alkalosis

74 The ABG analysis for a ABG for a 57 year-old patient reveals : p= 7.21, Electrolytes: Na 145 mEq/L, CI 105 mEq/L, HCO3 25 mEq/. The calculated anion gap is : أسئلة سابقة

a. 12
b.
c. 10
d. 15

75 Which of the following is a nursing measure directed for management of a post car accident patient who has respiratory alkalosis ? أسئلة سابقة

a. Ask if he is alright
b. Administer 02 therapy
c. Check body temperature
d. Ask him to breathe into a bag. Abed

76 Which of the following best defines pH? أسئلة سابقة

a. The amount of carbon dioxide in the blood
b. The concentration of free hydrogen ions
c. The level of oxygen saturation
d. The bicarbonate concentration in the blood

77 What is the normal pH range for arterial blood? أسئلة سابقة

a. 6.8 - 7.2
b. 7.0 - 7.4
c. 7.35 - 7.45
d. 7.5 - 7.8

78 The body maintains acid-base balance through which mechanisms? أسئلة سابقة

a. Respiratory system, urinary system, and liver function
b. Respiratory regulation, renal regulation, and chemical buffers
c. Liver function, chemical buffers, and cardiac function
d. Renal regulation, cardiac function, and respiratory system

79 Which of the following ABG values indicates respiratory acidosis? أسئلة سابقة

a. pH: 7.50, PaCO2: 30 mm Hg
b. pH: 7.30, PaCO2: 50 mm Hg
c. pH: 7.45, PaCO2: 35 mm Hg
d. pH: 7.35, PaCO2: 45 mm Hg

80 What is the primary cause of respiratory alkalosis? أسئلة سابقة

a. Hypoventilation
b. Hyperventilation
c. Excess bicarbonate
d. Excess acid intake

81 Which parameter indicates the amount of bicarbonate in the bloodstream? أسئلة سابقة

a. PaO2
b. SaO2
c. PaCO2
d. HCO3

أسئلة سابقة

82 A patient with a pH of 7.25 and HCO3 of 18 mEq/L likely has:

a. Metabolic alkalosis
b. Respiratory acidosis
c. Metabolic acidosis
d. Respiratory alkalosis

أسئلة سابقة

83 In an acid-base buffer system, a weak acid and its conjugate base maintain:

a. pH of urine
b. pH of blood and body fluids
c. pH of intracellular fluid
d. None of the above

أسئلة سابقة

84 The primary symptom of metabolic acidosis includes:

a. Hyperventilation
b. Light-headedness
c. Shallow breathing
d. Hyperactive reflexes

أسئلة سابقة

85 Which of the following is a normal PaCO2 range?

a. 22-26 mm Hg
b. 35-45 mm Hg
c. 45-55 mm Hg
d. 55-65 mm Hg

أسئلة سابقة

86 Compensation for respiratory acidosis primarily involves:

a. Increasing ventilation
b. Renal reabsorption of bicarbonate
c. Decreasing ventilation
d. Administration of IV fluids

أسئلة سابقة

87 A high anion gap is often indicative of:

a. Hypoalbuminemia
b. Gastrointestinal loss of bicarbonate
c. Ketoacidosis
d. Renal failure - Answer: C

أسئلة سابقة

88 What happens during hyperventilation?

a. CO2 levels decrease, leading to respiratory alkalosis
b. CO2 levels increase, leading to respiratory acidosis
c. O2 levels decrease, leading to metabolic acidosis
d. Bicarbonate levels decrease, leading to metabolic alkalosis

أسئلة سابقة

89 A patient with chronic obstructive pulmonary disease (COPD) is likely to have:

a. Respiratory alkalosis
b. Metabolic alkalosis
c. Respiratory acidosis
d. Metabolic acidosis

أسئلة سابقة

90 Which of the following conditions can cause metabolic alkalosis?

a. Prolonged vomiting
b. Diarrhea
c. Ketoacidosis
d. Renal failure

أسئلة سابقة

91 Which diagnostic test measures the partial pressure of oxygen in arterial blood?

a. SaO2
b. PaO2
c. HCO3
d. PaCO2

92 The pH of 7.48, PaCO2 of 32 mm Hg, and HCO3 of 24 mEq/L indicates: أسئلة سابقة

a. Uncompensated metabolic acidosis
b. Uncompensated respiratory alkalosis
c. Compensated metabolic alkalosis
d. Compensated respiratory acidosis

93 A decreased anion gap may be due to: أسئلة سابقة

a. Ketoacidosis
b. Hyperalbuminemia
c. Hypoalbuminemia
d. Metabolic acidosis

94 The condition characterized by pH > 7.45 and HCO3 > 26 mEq/L is: أسئلة سابقة

a. Respiratory acidosis
b. Respiratory alkalosis
c. Metabolic acidosis
d. Metabolic alkalosis

95 The most general effect of pH changes on the body is: أسئلة سابقة

a. Changes in oxygen levels
b. Changes in enzyme function
c. Changes in glucose levels
d. Changes in red blood cell count

96 Which symptom is commonly associated with respiratory acidosis? أسئلة سابقة

a. Restlessness
b. Hyperactive reflexes
c. Shallow breathing
d. Light-headedness

97 What does a bicarbonate level of 30 mEq/L suggest? أسئلة سابقة

a. Metabolic acidosis
b. Respiratory acidosis
c. Metabolic alkalosis
d. Normal bicarbonate level

98 Which of the following is NOT a chemical buffer system in the body? أسئلة سابقة

a. Bicarbonate buffer system
b. Phosphate buffer system
c. Protein buffer system
d. Oxygen buffer system

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 pH 7.30, PaCO2 55, HCO3 24 indicates: NCLEX

a. Uncompensated respiratory acidosis
b. Metabolic acidosis
c. Compensated respiratory alkalosis
d. Metabolic alkalosis

N2 Which condition causes metabolic alkalosis? NCLEX

a. Diarrhea
b. Prolonged vomiting / NG suction
c. DKA
d. Renal failure

N3 Select ALL causes of metabolic acidosis. SELECT ALL

a. Diabetic ketoacidosis
b. Severe diarrhea
c. Renal failure
d. Shock (lactic acidosis)
e. Hyperventilation

N4 **Normal anion gap is about:** NCLEX

a. 2–4 mEq/L
b. 8–12 mEq/L
c. 20–25 mEq/L
d. 30 mEq/L

N5 **The appropriate nursing action for a hyperventilating anxious patient with respiratory alkalosis is:** NCLEX

a. Give bicarbonate
b. Have him breathe slowly / into a bag
c. Increase FiO2
d. Give diuretic

N6 **Kussmaul's respiration is a compensatory mechanism for:** NCLEX

a. Respiratory acidosis
b. Metabolic acidosis
c. Respiratory alkalosis
d. Metabolic alkalosis

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

Normal pH	7.35–7.45.	**PaCO2**	35–45 mmHg (respiratory).
HCO3	22–26 mEq/L (metabolic).	**Ratio**	HCO3:H2CO3 = 20:1.
Resp acidosis	Hypoventilation, COPD, opioids.	**Resp alkalosis**	Hyperventilation, anxiety, pain, fever.
Met acidosis	DKA, diarrhea, renal failure, shock.	**Met alkalosis**	Vomiting, NG suction, diuretics.
Anion gap	Na − (Cl + HCO3); normal 8–12.	**Compensation**	Partial: pH abnormal; full: pH normal.
Mixed	PaCO2 and HCO3 both in the acid (or base) direction.	**Kussmaul**	Compensation for metabolic acidosis.

Answer Key

الإجابات

1. b — Uncompensated respiratory acidosis

2. b — 1,2,4

3. d — Metabolic acidosis
ناقصة في ورقة الامتحان والحل متظلل على قيمة PaCO2 respiratory acidosis. 2021 نفس السؤال في ميدترم إجابته (PaCO2 16) metabolic acidosis مع تعويض جزئي — HCO3 10 = سبب أيضي منخفض جداً.

4. a — pH 7.25, PaCO2 50 mmHg

5. c — Partially compensated respiratory alkalosis

6. c — Kussmaul's respiration

7. b — Partially compensated metabolic acidosis

8. a — pH 7.25, PaCO2 50 mmHg

9. c — Simultaneous respiratory and metabolic acidosis
ملاحظة: أسئلة أدوية القلب (nitroglycerin، dobutamine، noradrenaline، propofol، thrombolytics) في الميدترم ده اتشالت لأنها مش في محاضرات القائمة.

10. b — Partially compensated respiratory alkalosis

11. d — Partially compensated metabolic acidosis

12. a — pH 7.25, PaCO2 50 mmHg

13. c — Fully compensated metabolic acidosis

14. d — Mixed respiratory and metabolic acidosis

15. d — 15

16. d — Ask him to breathe into a bag

17. b — Uncompensated respiratory acidosis

18. b — Respiratory acidosis, partially compensated

19. c — Partially compensated metabolic acidosis

20. d — Respiratory alkalosis

21. a — Uncompensated respiratory acidosis

22. a — Uncompensated respiratory alkalosis

23. a — Partially compensated metabolic alkalosis

24. a — Fully compensated respiratory acidosis

25. a — Uncompensated respiratory acidosis

26. a — Mixed respiratory and metabolic acidosis

27. b — High anion gap metabolic acidosis

28. d — Compensated Respiratory alkalosis

29. b — Bicarbonate reabsorption is decreased

30. d — Partially Compensated Metabolic alkalosis Abed Critical Theory

31. c — Metabolic acidosis without compensation Respiratory acidosis without compensation

32. b — Uncompensated Respiratory Acidosis

33. d — Mixed Respiratory & Metabolic Acidosis

34. c — Fully Compensated Metabolic Acidosis Fully Compensated Respiratory Alkalosis

35. d — Instruct the client to breathe into a paper bag

36. c — Metabolic alkalosis

37. b — Potassium level of 3.0 mEq/L

38. a — Metabolic Acidosis Metabolic Alkalosis

39. d — Anion gap is decreased in ketoacidosis

40. c — PH: 7.25 - Paco2 50 mmhg

41. c — Metabolic Acidosis, Fully Compensated

42. c — Respiratory Acidosis Respiratory Alkalosis

43. c — Laryngeal obstruction

44. a — Metabolic acidosis

45. b — Uncompensated respiratory acidosis Simultaneous respiratory and metabolic acidosis

46. d — Radial Artery

47. d — High pCO2, high bicarbonate and High pH Abed Critical Theory

48. c — Metabolic Acidosis, Fully Compensated

49. a — Bicarbonate buffer

50. d — Hypoventilation (bradypnea) Abed Critical Theory

51. a — Metabolic Alkalosis, Uncompensated

52. d — Hyperaldosteronism

53. a — Usage of Diamox

54. c — Simultaneous respiratory and metabolic acidosis. Respiratory acidosis with complete renal compensation

55. b — Partially compensated respiratory alkalosis

56. d — Partially compensated metabolic acidosis

57. a — PH 7.25-PaCo2, 50 mmHg

58. b — Uncompensated Metabolic Alkalosis

59. c — The pH will rise and PCO2 will fall. The pH will fall and PCO2 will rise

60. b — Metabolic Alkalosis

61. d — The client is probably hyperventilating. Abed Critical Theory

62. a — Uncompensated Metabolic Alkalosis,

63. d —

64. a — Uncompensated Metabolic Alkalosis ,

65. c — 12

66. b — Uncompensated Metabolic Alkalosis

67. d — Fully Compensated Respiratory Acidosis

68. e — 1,2,4

69. b — Uncompensated Respiratory Acidosis

70. b — Respiratory Acidosis, Partially Compensated

71. a — Metabolic acidosis

72. c — Partially Compensated Metabolic Acidosis

73. d — Respiratory alkalosis

74. d — 15

75. d — Ask him to breathe into a bag. Abed

76. b — The concentration of free hydrogen ions

77. c — 7.35 - 7.45

78. b — Respiratory regulation, renal regulation, and chemical buffers

79. b — pH: 7.30, PaCO2: 50 mm Hg

80. b — Hyperventilation

81. d — HCO3

82. c — Metabolic acidosis

83. b — pH of blood and body fluids

84. a — Hyperventilation

85. b — 35-45 mm Hg

86. b — Renal reabsorption of bicarbonate

87. c — Ketoacidosis

88. a — CO2 levels decrease, leading to respiratory alkalosis

89. c — Respiratory acidosis

90. a — Prolonged vomiting

91. b — PaO2

92. b — Uncompensated respiratory alkalosis

93. c — Hypoalbuminemia

94. d — Metabolic alkalosis

95. b — Changes in enzyme function

96. a — Restlessness

97. c — Metabolic alkalosis

98. d — Oxygen buffer system

N1. a — Uncompensated respiratory acidosis
Low pH, high PaCO2, normal HCO3.

N2. b — Prolonged vomiting / NG suction
Loss of gastric HCl raises bicarbonate.

N3. a, b, c, d — Diabetic ketoacidosis; Severe diarrhea; Renal failure; Shock (lactic acidosis)
Hyperventilation causes respiratory alkalosis.

N4. b — 8–12 mEq/L
AG = Na − (Cl + HCO3).

N5. b — Have him breathe slowly / into a bag
Rebreathing CO2 corrects the low PaCO2.

N6. b — Metabolic acidosis
Deep rapid breaths blow off CO2.